

Software Design Document (SDD)

Project Title: File Backup and Sync Utility

1. Introduction

1.1 Purpose

The purpose of this project is to build a simple tool that simulates backing up files from one folder to another. It helps users understand how backup and syncing work by copying files and folders when changes are detected.

1.2 Scope

This is a beginner-level project that works with folders and files in an abstract way using Python. The app checks a source folder and copies new or updated files into a backup folder, simulating a basic sync operation.

2. System Overview

The system monitors a source folder and compares it with a backup folder. If it finds any new or modified files in the source folder, it copies them to the backup folder to keep both folders in sync.

3. System Features

3.1 Directory Comparison

- Compares the contents of two folders (source and backup).
- Detects new or changed files in the source folder.

3.2 File Copying

- Copies files from the source folder to the backup folder.
- Keeps the folder structure the same.

3.3 Simulation of Monitoring

- Runs in a loop or is manually triggered to check for updates.
- Simulates real-time syncing by checking periodically or on command.

3.4 Log Messages

- Prints actions like "Copied file.txt to backup/" or "No changes found".
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4. Optional Ideas

- Add timestamps to copied files or logs.
- Allow deleting files from the backup if they're removed in the source.
- Let the user select folders using input.
- Display a summary report at the end of the sync.
- Add a “dry run” mode that shows what would happen without copying.